

OTT Streaming Platform Business Insights and Recommendations

1. Business Objective

The objective of the project was to analyze OTT streaming platform viewing patterns, content performance, user behavior, subscription and churn patterns, ratings and feedback, and platform engagement to identify factors affecting viewing engagement and customer retention.

The analysis aimed to:

- Identify high-performing and low-performing content.
- Understand viewing behavior across users, regions, age groups and devices.
- Analyze content performance by content type, genre, title and platform.
- Understand viewing patterns across different times of the day.
- Evaluate subscription and churn patterns.
- Analyze ratings and customer feedback.
- Identify platform and regional engagement patterns.
- Analyze relationships between viewing duration and completion percentage.
- Provide actionable recommendations for improving content engagement and customer retention.

2. Key Findings / Business Insights

Analysis Finding	Business Insight	Business Impact
Total Watch Hours fluctuated across the years , with a decline in 2023, recovery in 2024 and another decline in 2025.	Overall engagement has not followed a consistent upward trend.	Monitoring yearly engagement can help identify periods of declining viewing activity and support timely corrective action.
44.4% of users were churned, while 55.6% were retained.	Customer retention is a significant business challenge for the OTT platform.	Reducing churn can help improve the stability of the active customer base.
Average completion percentage was approximately 52.36% and varied across age groups and subscription types.	Engagement differs across customer segments, although the differences are relatively small.	Segment-level content recommendations can help improve engagement.

Analysis Finding	Business Insight	Business Impact
Content releases increased substantially in later release years.	The content catalogue has expanded considerably over time.	Catalogue expansion should be evaluated based on actual viewing engagement rather than content volume alone.
NetStream generated the highest watch hours and views among the platforms , followed by AmazePrime, while DizPlay+ contributed considerably less.	Platform engagement is uneven, with NetStream contributing the strongest viewing activity.	Strong platform-content combinations can be used as benchmarks, while lower-performing platforms require further investigation.
Joker recorded the highest number of views among the displayed top titles , followed by several titles with relatively high viewing activity.	Viewing activity is concentrated among certain titles.	High-performing titles can receive greater visibility and can provide characteristics for content recommendation strategies.
East and West generated relatively higher watch hours and views , while South recorded comparatively lower engagement.	Viewing behavior varies geographically.	Regional content and promotional strategies can be adjusted according to engagement patterns.
Smart TV had the highest number of users , followed by Mobile, Tablet and Laptop.	Smart TV and mobile devices represent important access points for the platform.	Maintaining a consistent and convenient viewing experience across major devices can support engagement.
Middle Age users generated the highest watch hours among the displayed age categories , while Young Adults generated comparatively lower watch hours.	Viewing engagement differs across age segments.	Age-specific content recommendations can help address differences in viewing behavior.
Smart TV generated the highest total watch hours among the device categories in the age-device analysis.	Device preference differs across user segments.	Platform experience and recommendations can be optimized according to device usage patterns.
Movies generated approximately 72.87% of total views , while TV Shows generated approximately 27.13%.	The available viewing activity is strongly movie-led by view count.	Movie content should remain an important part of content promotion, while TV Show discovery and engagement can be investigated.
TV Shows generated approximately 73K watch hours	Although Movies generate more viewing records, TV	Content strategy should consider both view volume and watch-

Analysis Finding	Business Insight	Business Impact
compared with approximately 18K for Movies.	Shows generate substantially more total watch time.	hour contribution rather than relying on views alone.
Night recorded the highest watch hours , followed by Evening, Afternoon and Morning.	Users show stronger viewing activity during Night and Evening periods.	Promotions and personalized recommendations can be scheduled around high-engagement periods.
The relationship between Watch Duration and Completion Percentage was weak.	Longer viewing duration alone does not strongly explain completion behavior.	Content characteristics and user preferences should be considered alongside watch duration when improving completion.
Average completion percentages were relatively similar across devices , with Tablet and Laptop slightly higher than Smart TV and Mobile.	Device type does not show large differences in completion performance.	Content and recommendation improvements may have greater value than focusing only on device-specific completion differences.
Free Trial had the highest number of churned users (483) , followed by Basic (280) and Premium (125).	Churn is concentrated in particular subscription groups, with Free Trial showing the highest churn count.	Trial-to-paid conversion and subscription retention strategies should receive particular attention.
Churned users had approximately 184 average watch minutes compared with approximately 178 minutes for retained users.	Higher watch duration alone does not clearly distinguish retained and churned users.	Retention strategies should consider subscription and renewal factors rather than relying only on viewing duration.
Subscription type showed an association with churn in the analysis.	Customer retention behavior differs across subscription types.	Subscription-specific retention strategies can be developed.
Renewal status showed a strong association with churn.	Renewal behavior is an important indicator of customer retention.	Renewal-stage interventions and targeted retention communication should be prioritized.
Device type did not show a significant relationship with churn in the analysis.	Device usage alone does not appear to be an important churn discriminator.	Churn strategies should focus more on subscription, renewal and engagement factors than device type.
Positive feedback represented approximately 40.30% of feedback, while negative	Customer feedback is mixed, with positive and negative	Negative feedback should be investigated to identify content

Analysis Finding	Business Insight	Business Impact
feedback represented approximately 39.42%.	feedback at relatively similar levels.	or platform areas requiring improvement.
Average rating was approximately 3.01.	Overall content/customer rating performance is moderate.	Content quality and audience satisfaction should be monitored together with viewing performance.
Some titles had very high average ratings but relatively low viewing volume.	High-quality or highly rated content may not automatically receive high visibility.	Highly rated, under-watched titles can be considered for targeted promotion and recommendation placement.
High watch-hour titles did not always correspond to the highest ratings.	Viewing popularity and rating performance represent different aspects of content success.	Content decisions should consider both engagement and audience satisfaction.
Average popularity score differed between Movies and TV Shows, with TV Shows showing the higher average popularity score.	Content type differs in its average popularity performance.	Popularity indicators can be combined with viewing behavior when evaluating content performance.
The content-country analysis contained a large Unknown category, while the United States was among the major identified countries.	Country metadata is incomplete for a significant portion of the content catalogue.	Improving content metadata can strengthen geographical content analysis and regional decision-making.
The platform-genre analysis showed NetStream dominating total watch hours across several genres.	NetStream has strong engagement across multiple content categories rather than depending on only one genre.	Successful platform-genre combinations can guide content promotion and recommendation strategies.
The decomposition analysis showed that watch hours vary across combinations of Region, Age Category, Platform and Content Type.	OTT engagement is influenced by combinations of user and content characteristics rather than a single factor.	Multi-dimensional segmentation can support more targeted content and engagement strategies.
The dashboard's Q&A analysis identified Drama as the answer to the question "Which genre has the highest rating?"	Interactive exploration can help users investigate specific business questions quickly.	Management users can use interactive analysis to explore content performance from different perspectives.

3. Recommendations

Recommendation Table

Insight	Recommendation	Expected Benefit
Yearly watch hours fluctuate.	Monitor yearly viewing trends and investigate the content, platform and user segments responsible for declines.	More timely identification of engagement declines and better content planning.
Churn rate is 44.4%.	Develop targeted retention strategies for users approaching renewal or showing retention risk.	Improved customer retention and reduced churn.
Free Trial has the highest number of churned users.	Strengthen trial-to-paid conversion using timely communication, relevant content recommendations and suitable subscription offers.	Improved conversion of trial users into retained subscribers.
Subscription type is associated with churn.	Evaluate retention strategies separately for Free Trial, Basic and Premium users.	More relevant subscription-specific retention actions.
Renewal status is strongly associated with churn.	Focus retention communication around renewal periods and identify users who do not renew.	Improved renewal performance and reduced customer loss.
Watch duration alone does not clearly distinguish churned and retained users.	Combine viewing behavior with subscription, renewal and content-preference information when identifying retention opportunities.	Better identification of customer retention patterns.
Movies generate 72.87% of views.	Maintain strong visibility for high-performing movie content and analyze the genres and titles driving movie demand.	Continued high viewing activity and better content promotion decisions.
TV Shows generate approximately 73K watch hours despite fewer views.	Investigate successful TV Show characteristics and promote high-engagement series more effectively.	Increased discovery and engagement with TV Show content.
NetStream has the highest watch-hour contribution.	Analyze successful NetStream content and platform-genre combinations and use the findings to guide content strategy.	Better replication of successful engagement patterns.
DizPlay+ contributes considerably fewer watch hours and views.	Investigate content availability, genre mix and title performance on lower-performing platforms.	Improved platform engagement.

Insight	Recommendation	Expected Benefit
Certain titles receive substantially more views.	Give high-performing titles greater visibility through recommendations, featured sections and promotional placement.	Higher viewing activity from proven content.
Some highly rated titles have low viewing volume.	Promote highly rated under-watched titles through personalized recommendations and targeted placement.	Better utilization of high-quality content.
Ratings and watch hours do not always move together.	Evaluate content using multiple indicators such as views, watch hours, ratings and popularity rather than one metric alone.	More balanced content investment decisions.
Positive and negative feedback are both substantial.	Analyze negative feedback by title, genre, platform and content type to identify areas requiring improvement.	Improved customer satisfaction and content experience.
Night and Evening have the highest watch-hour activity.	Schedule important recommendations, promotions and engagement campaigns during high-viewing periods.	Better engagement with promotional content.
Regional engagement varies.	Develop region-specific content recommendations and promotional strategies based on viewing patterns.	Improved regional engagement.
Young Adults show comparatively lower watch-hour contribution.	Investigate content preferences of Young Adults and improve recommendations for this segment.	Increased engagement among lower-performing age segments.
Smart TV has the highest number of users.	Continue prioritizing a smooth viewing experience on Smart TV while maintaining consistent experiences across all devices.	Improved accessibility and user experience.
Device completion differences are relatively small.	Avoid focusing retention strategy primarily on device type; prioritize content and customer behavior factors.	More effective allocation of analytical and product efforts.
Content catalogue has expanded substantially in later years.	Evaluate new content based on actual views, watch hours, ratings and popularity rather than catalogue size alone.	Better content investment decisions.
Country information contains a large Unknown category.	Improve completeness and consistency of country metadata.	More reliable geographical and content-market analysis.

Insight	Recommendation	Expected Benefit
Engagement varies across Region × Age Category × Platform × Content Type combinations.	Use multi-dimensional segmentation to create more targeted recommendations and promotional strategies.	More personalized customer engagement.

4. Overall Recommendation

The OTT platform should focus on **content engagement and customer retention together** rather than optimizing only for the number of views.

High-performing titles and platforms should receive greater visibility, while highly rated but under-watched content should be identified for targeted promotion. Since **Movies generate the majority of views but TV Shows generate substantially higher watch hours**, content decisions should consider both viewing volume and depth of engagement.

The platform should also address the **44.4% churn rate** by focusing on subscription-specific and renewal-stage retention strategies. Free Trial users require particular attention because they account for the highest number of churned users.

Regional, age-group and platform differences should be used to improve personalized recommendations. Since Night and Evening show stronger viewing activity, engagement campaigns and recommendations can also be aligned with these high-viewing periods.

Finally, content decisions should use a combination of **views, watch hours, ratings, popularity, completion and user behavior**, rather than relying on a single KPI.

5. Conclusion

The project successfully analyzed the OTT streaming dataset from multiple perspectives, including **user behavior, content performance, viewing patterns, platform performance, subscription and churn, ratings and feedback, regional engagement, device usage and time-of-day viewing behavior**.

The analysis identified several important business patterns. **NetStream generated the highest watch-hour contribution, Movies generated the majority of views, while TV Shows generated substantially higher total watch hours**. Certain titles attracted considerably higher viewing activity, indicating that content performance is concentrated among selected titles.

The analysis also identified a significant **44.4% churn rate**, with Free Trial users having the highest number of churned users. Subscription type and renewal status are important areas for retention analysis, while device type does not show a significant relationship with churn.

Regional and demographic analysis showed differences in viewing engagement, while Night and Evening emerged as important viewing periods. The ratings and feedback analysis showed mixed customer sentiment, indicating opportunities to investigate negative feedback and promote highly rated but under-watched content.

The project objectives were therefore achieved by converting OTT viewing data into **business insights, actionable recommendations and data-driven content and retention strategies**.

Overall Business Implication

The OTT platform can improve performance by combining:

High-performing content focus + personalized recommendations + subscription retention + renewal optimization + regional targeting + platform optimization + engagement-timing strategies.

Final Conclusion

The project successfully analyzed OTT viewing patterns, identified key content, user, platform and retention patterns, evaluated important relationships within the data, and provided actionable recommendations to support content decisions, improve viewer engagement and strengthen customer retention.

6. Model / Statistical Analysis

Statistical Analysis

The OTT project used statistical analysis to validate selected business assumptions related to **churn, renewal, user behavior and viewing engagement**.

The analysis included:

- **One-Sample Proportion Z-Test**
- **Independent Two-Sample T-Test**
- **Chi-Square Test of Independence**
- **Simple Linear Regression**

The statistical analysis helped determine whether observed patterns were statistically meaningful rather than relying only on visual observations.

Key Statistical Findings

Analysis	Finding	Business Interpretation
Churn proportion vs 20% benchmark	Churn was significantly different from the 20% benchmark.	The observed churn level represents a significant retention concern.
Renewal proportion vs 70% benchmark	Renewal proportion was significantly different from the 70% benchmark and was below the benchmark.	Renewal performance requires attention.
Watch Duration vs Churn	No statistically significant difference was found.	Watch duration alone should not be used as a primary churn indicator.
Completion Percentage vs Churn	No statistically significant difference was found.	Completion percentage alone does not clearly distinguish churned and retained users.

Analysis	Finding	Business Interpretation
Subscription Type vs Churn	Significant association was found.	Retention behavior differs across subscription types.
Renewal Status vs Churn	Strong significant association was found.	Renewal behavior is strongly related to customer retention status.
Device Type vs Churn	No statistically significant association was found.	Device type should not be treated as a major churn driver.
Watch Duration vs Completion Percentage	A statistically significant but weak positive relationship was found.	Longer watch duration is associated with slightly higher completion, but it has limited predictive power.

The notebook's statistical summary reports the above hypothesis-test results, including the significant associations for subscription type and renewal status and the non-significant relationships for device type, watch duration vs churn, and completion vs churn.

Model Analysis

Simple Linear Regression

The project used **Simple Linear Regression** to examine whether watch duration could predict completion percentage.

Independent Variable: Watch Duration in Minutes

Dependent Variable: Completion Percentage

The model produced:

- **R = 0.2205**
- **R² = 4.86%**
- **p < 0.001**
- **Test MAE = 23.74**
- **Test RMSE = 27.50**

The regression therefore indicates a **statistically significant but weak positive relationship** between watch duration and completion percentage. The low R² indicates that watch duration alone explains only a small portion of the variation in completion percentage.

The business implication is that completion performance should also consider factors such as **content genre, content duration, user preferences, device, viewing time, ratings and engagement behavior**, rather than relying only on watch duration.

7. Challenges and Limitations

Data Limitations

1. Content Metadata Completeness

The country/content-type analysis contains a large **Unknown** category.

Incomplete country information can limit the ability to perform detailed geographical content analysis.

2. Extreme Viewing Values

The viewing data contains highly variable watch-duration values, including extremely high viewing durations.

Such extreme values can influence averages and require careful interpretation.

3. Multiple Platform Usage

Users can appear across multiple platforms.

Therefore, platform-level churn should not be interpreted as users specifically **churning from a particular platform**.

Platform analysis should instead be interpreted as **churned users associated with platform usage**.

4. Available Variables

The analysis is limited to the information available in the OTT dataset and dashboard.

Factors such as:

- Competitor activity
- Advertising expenditure
- External marketing campaigns
- Customer income
- Detailed browsing behavior
- External content availability
- Competitor catalogue

were not included in the available analysis.

5. Historical Data

The available viewing period covers the years shown in the dashboard and may not fully capture longer-term changes in viewing behavior, market conditions or future trends.

8. Analysis Limitations

- The analysis is based only on the variables available in the OTT dataset.
- Correlation and statistical association do not imply causation.
- Dashboard visual patterns should not automatically be interpreted as causal relationships.
- Key Influencers identify associations in the analyzed data and should not be treated as proof of causality.

- Watch duration has limited predictive power for completion percentage.
 - Device type was not statistically associated with churn in the analysis.
 - Churn counts by subscription type should not automatically be interpreted as churn rates because subscription groups have different numbers of users.
 - Platform-level churn should not be interpreted as users churning specifically from an individual platform because users may use multiple platforms.
 - Some content metadata, particularly country information, contains Unknown values.
 - Ratings, views, watch hours and popularity measure different aspects of content performance and should be interpreted together.
 - The analysis does not establish that one business factor directly causes another.
 - The regression model uses only watch duration as the predictor and therefore has limited predictive capability.
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9. Future Scope

Future Improvements

1. Real-Time OTT Dashboard

The existing dashboard can be enhanced to monitor OTT KPIs in near real time, including:

- Total Users
- Total Views
- Total Watch Hours
- Average Watch Duration
- Average Completion
- Churn Rate
- Retention Rate
- Average Rating
- Positive Feedback %
- Platform Performance

2. Advanced Customer Segmentation

Future analysis could create more detailed user segments such as:

- Highly Engaged Users
- Low-Engagement Users
- At-Risk Users

- Trial Users
- Highly Retained Users
- High-Viewing Users

These segments could be used for personalized content recommendations and retention strategies.

3. Improved Churn Prediction

Future models could incorporate multiple variables such as:

- Viewing frequency
- Watch duration
- Completion percentage
- Subscription type
- Renewal behavior
- Ratings
- Rewatch behavior
- Platform usage
- Time of day
- Content preferences

This could improve churn-risk identification compared with using individual variables alone.

4. Advanced Content Recommendation System

A recommendation system could be developed using:

- User viewing history
- Genre preferences
- Ratings
- Completion behavior
- Rewatch behavior
- Platform usage
- Age segment
- Viewing time

This could provide more personalized content recommendations.

5. Content Performance Prediction

Future machine-learning models could be introduced to predict:

- Expected views

- Expected watch hours
- Content completion
- Content popularity
- Potential high-performing titles

6. Regional Content Strategy

Future analysis could combine viewing patterns with more complete country and regional metadata to identify:

- Regional content preferences
- High-performing genres by region
- Regional platform preferences
- Regional engagement opportunities

7. Time-Based Engagement Analysis

More detailed time-based analysis could examine:

- Day of week
- Weekend vs weekday
- Monthly trends
- Seasonal viewing
- Peak viewing hours

This could support more effective content releases and promotional scheduling.

8. Advanced Feedback Analysis

Future analysis could examine feedback at a deeper level by combining:

- Rating
- Feedback category
- Genre
- Title
- Platform
- Completion percentage
- Watch duration

This could help identify why certain content receives negative or positive audience responses.

Final Business Summary

The OTT analysis shows that **content engagement, customer retention and user segmentation should be considered together.**

The strongest opportunities identified from the dashboard are:

1. Reduce the 44.4% churn rate.
2. Improve Free Trial-to-retained-user conversion.
3. Continue leveraging high-performing movie content.
4. Investigate the strong watch-hour contribution of TV Shows.
5. Promote highly rated but under-watched titles.
6. Use regional and age-group differences for personalization.
7. Focus engagement campaigns around Night and Evening viewing periods.
8. Use multiple content-performance metrics rather than relying only on views or ratings.
9. Improve metadata completeness, particularly the Unknown country values.
10. Use subscription and renewal behavior as important areas for retention strategy